

- k) All seams of the silencer shall be sealed with a duct sealing compound and shall be airtight when operating at 2000 Pa static pressure and rated capacity. Silencers shall not fail structurally when subjected to a differential air pressure of 2000 Pa.
- l) The pressure drops through the attenuator selected shall not exceed nominal design pressure drop, and dynamic insertion loss of the attenuator. The Contractor shall also carry out noise calculations and select sound attenuators with higher insertion losses, if necessary, to comply with the specified noise criteria, without additional cost.
- m) The silencers for exhaust fans used for smoke purging/exhaust shall be structurally sound to withstand an air stream temperature of 250°C for at least two hours. Test certificate and test report shall be submitted for notice of no objection.
- n) The sound attenuators shall be delivered with blocked ends to avoid ingress of rubble equipment and shall be checked for damage during delivery to site, before installation and after installation.
- o) The contractor shall provide structural support of hot dip galvanised steel for installation of attenuators where necessary.
- p) For the BOQ unit in Square Meter, the sound attenuator area shall be measured based on the external surface area of the ducts. The external surface area shall be calculated by measuring the overall perimeter (width and depth) of the sound attenuator and multiplying it by the overall length from flange face to flange face of the attenuator. The Sq. Mtr line item shall be used when the attenuator size is not available in the BOQ, with NONO from the Engineer/Employer.

10.3.2 Acoustic Linings for Duct, Walls, Slabs, Ceilings, Beams and Columns

- a) Acoustic linings on the walls, floor slabs and ceilings beam, and columns of the VAC equipment rooms, air plenums and ventilation shafts shall be applied where required to meet the noise level limits.
- b) Linings shall be protected with a suitable material to reduce abrasion loss. The material and fixing details shall be submitted to Engineer for issue of notice of no objection. Wall linings shall be protected with perforated (min. 30%) aluminium sheet of minimum 0.8 mm thickness.
- c) Linings shall be 50 mm thick rockwool insulation with 105 kg/m³ and shall have the following sound absorption coefficients according to ASTM C423-66.
- d) **Acoustic Lining:** Providing and Fixing acoustic lining in plant room as per specifications and drawings.

Octave Bands	Coefficient
125	0.31
250	0.81
500	1.16
1000	1.09
2000	1.06
4000	1.13
NCR	1.05

- e) Sufficient 'C' and 'Z' channels, which shall be hot-dipped galvanised, shall be provided to hold the acoustic insulation materials and perforated galvanised steel sheet.
- f) Acoustic Lining for VAC Rooms: In addition to the above requirements for acoustic lining for walls, slabs, ceilings, beams and columns acoustic linings for VAC Rooms shall be applied where required to meet the noise level limits.

- q) Requirement of acoustic enclosures for fans or acoustic lining of plant room walls shall be avoided, unless the fan casing breakout sound power levels demand such an arrangement for lowering the plantroom break-out noise to specified criteria. The same shall be installed by the contractor as per the cost quoted in the provisional BOQ. The contractor shall be responsible for the design and installation of the acoustic enclosure and wall-lining including the support framework based on manufacturer's recommended installation details.
- r) Any acoustic enclosure/wall-lining installed shall be 2 hours fire-rated at 250oC. The acoustic lining material shall be rockwool and its thickness shall be as per the requirement, but not lower than 50mm. The lining shall be covered by 0.8 mm thick perforated aluminium sheet. The contractor shall also substantiate the addition of acoustic lining will not affect the fire resisting period and integrity of the fire-rated enclosure and the plantroom.
- s) Adequate amount of access panel shall be allowed by the contractor at the acoustic enclosure of fans such that all the equipment and accessories is fully accessible.
- t) Acoustic lining for shafts/Ducts shall be installed by contractor when the actual sound power levels at external or public areas is higher than the calculated values with the selected attenuators. The same shall be installed by the contractor as per the cost quoted in the provisional BOQ. The contractor shall be responsible for the design and installation of the acoustic lining including the support framework based on manufacturer's recommended installation details. The acoustic lining shall be fire-rated for 2hours at 250oC.
- g) Installation
 - i. Ensure that the surface substrate is Clean, dry and at ambient temperature. This involves removing dust, scale and flaking paint.
 - ii. Mechanical Fastening is recommended for all walls/ under slab installation. Fastening system should be suitable to the civil structure.
 - iii. It is advisable to pin near the edges of the slab at 600mm intervals and across the slab at 600mm square interval.
 - iv. Material should be able to drill or cut to accommodate the installation of the other services after the insulation is fixed.
 - v. Factory applied adhesive or wet adhesive can be used to improve the bond between the foam and the wall/ceiling.
 - vi. Acoustic lining of wall shall be terminated approximately 15cm above the finished floor to prevent damage to insulation due to accidental water logging in AHU/ plat room.
 - vii. Use 50mm or more width aluminium file tape.

10.4 Factory Visit and Factory Acceptance Test (FAT)

- a) This section shall comply with the requirements specified in "Section 1.3" of the technical requirements. Clause wise compliance matrix for the shall be submitted for notice of no objection.

10.5 Submittal for Vendor / Material Technical Submittal

- a) This section shall comply with the requirements specified in "Section 1.2" of the Technical Requirements. Clause wise compliance matrix for the same shall be submitted for notice of no objection.
- b) Complete catalogue information and shop drawings for material, isolators and equipment and their method of mounting shall be submitted to Engineer for issuance of NONO.
- c) Sound attenuators selection with verification of sound attenuation calculation shall be submitted.
- d) Shop drawings showing the make and model number of the sound attenuators, acoustic enclosure, inertia block, etc. shall be submitted.
- e) Sound Performance test data and performance of the silencer and VAC equipment by accepted and recognized testing authority shall be submitted.

10.6 Installation Requirements:

- a) Silencers shall be installed in accordance with the manufacturer's recommendations and the same shall be submitted to Engineer for notice of no objection.